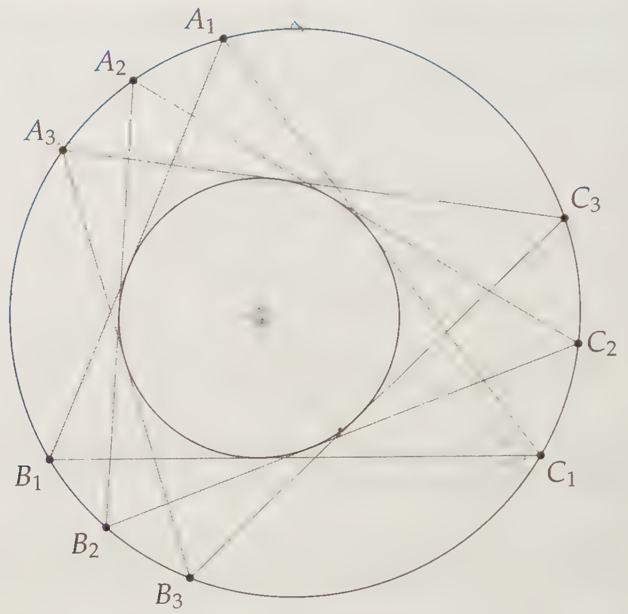


Poncelet's Porism

Man's mind, stretched by a new idea, never goes back to its original dimensions. – Oliver Wendell Holmes

CHAPTER 14 Three-Dimensional Geometry

So far in this book we've confined ourselves to zero dimensions (points), one dimension (lines and line segments), or two dimensions (basically everything else in the book). However, we all know the world around us seems three-dimensional – that in addition to left and right, and forwards and backwards, there's up and down.

In this chapter we step off the page and into the 'real world' by considering three-dimensional objects. As we'll see, this isn't much of a step; much of our work in three-dimensional geometry will consist of reducing problems to two-dimensional situations we already know how to handle.

14.1 Planes

As mentioned in Section 1.2, a **plane** is a 'flat' two-dimensional surface that extends forever. For example, when your book is completely flat, this page is part of a plane. In this section, we'll explore planes, as well as how lines and planes can intersect in space.

Problems

Problem 14.1: We know that given any two points, we can draw exactly one line through the two points.

(a) Given two points, how many different planes pass through the two points?

(b) Given three points, is it always possible to find a plane that passes through all three points?

(c) Given any four points, is it always possible to find a plane that passes through all four points?

Problem 14.2:

(a) When two lines intersect, their intersection is a point. When two planes intersect, what kind of figure is their intersection?

(b) Is it possible for two planes to never intersect?

(c) Can there be a line and a plane that never intersect?

(d) Remember that two lines are parallel if they are *in the same plane* but do not intersect. Can two lines in space be such that they are not parallel but still do not intersect?

(e) Given any two intersecting lines, is there always a plane that contains both lines?

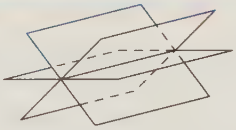
Problem 14.3:

(a) How can we tell if a line is perpendicular to a plane? Give an example in the 'real world' of a line and a plane that are perpendicular.

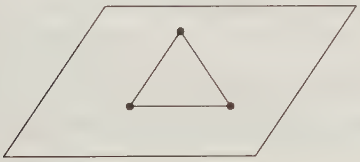
(b) How can we tell if two planes are perpendicular? Give an example in the 'real world' of two planes that are perpendicular.

We start with a fundamental question about planes.

Problem 14.1: How many points are needed to determine a unique plane?



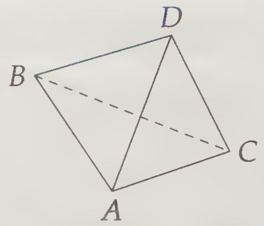
Solution for Problem 14.1: If we only have two points, we can find a line through them, but there are infinitely many planes that can pass through this line (and therefore pass through our initial two points). An example is shown in the figure at right, in which three planes are shown passing through the same line.



Given three or more collinear points (meaning they are all on the same line), we can still find infinitely many planes that pass through all three points, but if we have three points that are not collinear, then we have a triangle. The plane of this triangle is the unique plane that passes through the three points, as shown at left.

Sidenote: Why do most telescopes have three legs? Why do photographers use a tripod, which has three legs, to stabilize their cameras instead of something with four legs? The answer: three points determine a plane, so wherever you place a stand with three legs, it will be stable.

While any three points are **coplanar**, meaning there is a plane through all three of them, it is easy to find a set of four points that are **noncoplanar**. Three vertices of a triangle and a point not in the plane of the triangle give us an example in which no plane passes through four given points. Such a situation is depicted at right, where $\overline{BC}$ is dashed because it is in the 'back' of the figure. (We wouldn't be able to see $\overline{BC}$ if this figure were rendered in three dimensions.) □



Important: Three noncollinear points determine a plane.

Now we consider how lines and planes in space can be related.

Problem 14.2:

(a) When two lines intersect, their intersection is a point. When two planes intersect, what kind of figure is their intersection?

(b) Is it possible for two planes to never intersect?

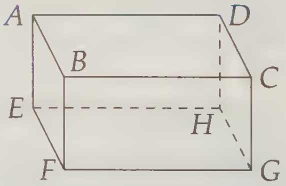
(c) Can there be a line and a plane that never intersect?

(d) Remember that two lines are parallel if they are *in the same plane* but do not intersect. Can two lines in space be such that they are not parallel but still do not intersect?

(e) Given any two intersecting lines, is there always a plane that contains both lines?

Solution for Problem 14.2:

(a) Consider a standard rectangular room as shown at right, where $EFGH$ is the 'floor' and $ABCD$ is the 'ceiling'. The walls are portions of planes. Look at where a wall meets the floor, and you'll see an example of two planes intersecting to form a line. Similarly, any two planes that intersect form a line.



(b) In a typical room, the ceiling is a constant distance from the floor. This means that if the ceiling and the floor extended forever in all directions, they would be planes that never intersect. Such planes are called **parallel**, just as lines in the same plane that never intersect are called parallel. For example, in our figure above, $ABCD$ and $EFGH$ are parallel planes.

(c) The intersection of a wall and the ceiling (line $\overleftrightarrow{AB}$, for example) will never meet the floor (plane $EFGH$). This is an example of a line that is parallel to a plane.

(d) Lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CG}$ do not intersect and are not in the same plane. Since they are not coplanar, they are not parallel. But they still don't intersect! We call noncoplanar lines that do not intersect **skew lines**.

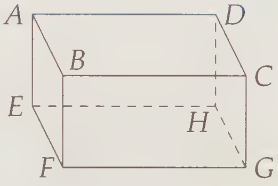
(e) Yes. Let X be the intersection point of the two lines, Y be a different point on one of the lines, and Z a different point on the other line. These three points determine a plane. Clearly, both $\overleftrightarrow{XY}$ and $\overleftrightarrow{XZ}$ are in the plane because $\overline{XY}$ and $\overline{XZ}$ are.

□

After dealing with parallels, you know what's next.

Problem 14.3:
(a) How can we tell if a line is perpendicular to a plane? Give an example in the 'real world' of a line and a plane that are perpendicular.
(b) How can we tell if two planes are perpendicular? Give an example in the 'real world' of two planes that are perpendicular.

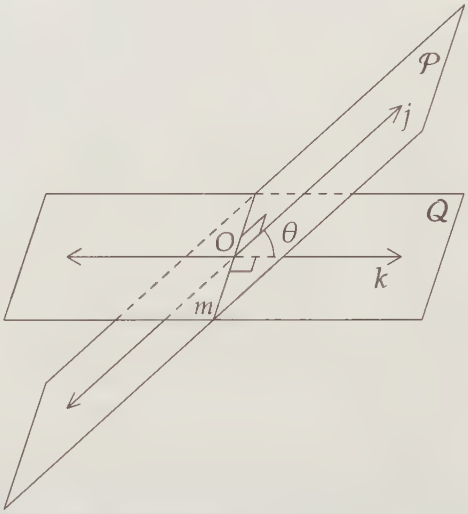
Solution for Problem 14.3: We'll use our trusty room for our 'real world' examples.



(a) Suppose line k intersects plane $\mathcal{P}$ at point X. If k is perpendicular to every line in $\mathcal{P}$ that passes through X, then we say that line k is perpendicular to the plane. The line of intersection between two adjacent walls in a typical room is perpendicular to the ceiling and the floor of the room. For example, $\overleftrightarrow{AE}$ is perpendicular to $ABCD$.

(b) Suppose planes $\mathcal{P}$ and $\mathcal{Q}$ intersect in line m, and let O be a point on line m. Consider lines j and k in $\mathcal{P}$ and $\mathcal{Q}$, respectively, such that each of j and k is perpendicular to m at point O. The angle between j and k, which is marked with a θ in the diagram, is the angle between the planes. We call such an angle between planes a **dihedral angle**. If this angle is a right angle, the planes are perpendicular.

In a typical room, a wall is perpendicular to the ceiling and to the floor. For example, plane $ABCD$ is perpendicular to plane $BCGF$ in our 'room' above.

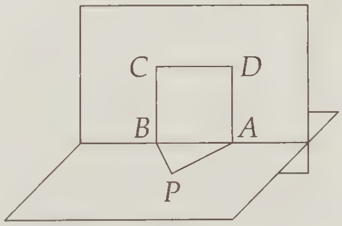


□

Exercises

14.1.1 Line m is perpendicular to plane $\mathcal{P}$ at point X. Line k is in plane $\mathcal{P}$ and passes through X. Are lines m and k necessarily perpendicular?

14.1.2 Triangle PAB and square $ABCD$ are in perpendicular planes at right. Given that $PA = 3$, $PB = 4$, and $AB = 5$, what is PD? *(Source: AMC 12)*



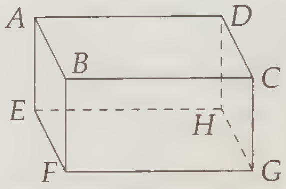
14.1.3 Is it true that the nearest point on line m in space to a point P in space is the foot of the perpendicular segment from P to m? Why or why not?

14.1.4 Is it true that the nearest point on plane $\mathcal{P}$ in space to a point X in space is the foot of the perpendicular segment from X to plane $\mathcal{P}$? Why or why not?

14.1.5 Two planes, $\mathcal{M}$ and $\mathcal{N}$ are each perpendicular to a third plane, $\mathcal{P}$. $\mathcal{M}$ and $\mathcal{N}$ intersect in line m. Is line m perpendicular to $\mathcal{P}$? (No proof necessary.)

14.2 Prisms

We move now from considering one- and two-dimensional figures in space to thinking about solid figures. Our 'room' was so useful in the previous section that we'll start with it here.



Mathematically speaking, $ABCDEFGH$ at the right is a **right rectangular prism**. Since all the boundaries of $ABCDEFGH$ are polygons, $ABCDEFGH$ can also be called a **polyhedron**. Each of these boundary polygons is a **face** of the polyhedron. The 'rectangular' part of 'right rectangular prism' refers to the fact that the top and bottom faces ($ABCD$ and $EFGH$) are rectangles. We typically call the 'top' and 'bottom' the **bases** of the prism. The 'right' part refers to the fact that all the edges connecting the two bases meet the bases at right angles.

Which leaves us with the 'prism' part. A **prism** is a three-dimensional solid figure with two congruent parallel faces, and with parallelograms as the other faces. The faces that are not bases are sometimes called the **sides** of the prism. In a **right prism** all of these sides are rectangles. As already noted, the congruent parallel faces are the bases of the prism. Hence, any pair of opposite sides of $ABCDEFGH$ above could be considered the bases of the prism.

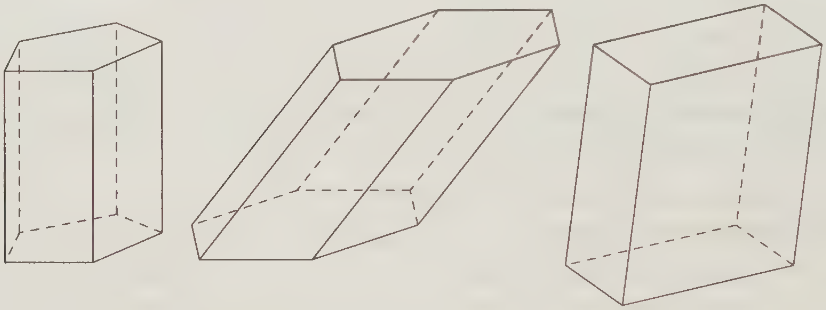


Figure 14.1: Some Prisms

Figure 14.1 shows some more prisms. The first is a right regular pentagonal prism because the base is a regular pentagon and the segments connecting the bases are perpendicular to the bases. The second is a hexagonal prism because the bases are hexagons (note that we don't call this one a 'right hexagonal prism'). The third has parallelograms as its bases, as well as all its sides. Such a prism is given the special name **parallelepiped**. As one last bit of vocabulary, if there's a 'regular' in the description of a prism, it means the base is a regular polygon.

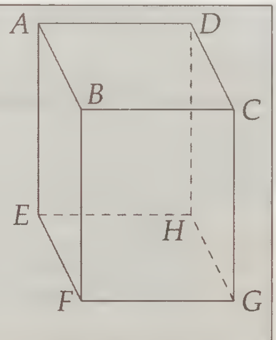
Make sure you understand what a prism is, but you don't have to worry about all those other names. One reason you don't have to worry too much is that they aren't used consistently. For example, the 'right' is often left out of the description, and you're meant to infer from the problem that the prism is indeed a right prism. Sometimes in this book we'll do this, such as by calling our original prism $ABCDEFGH$ a 'rectangular prism'.

Unfortunately, the vocabulary lesson isn't over yet. Just as we measure the region contained in a two-dimensional figure with area, we can measure the space inside a three-dimensional figure with **volume**. We do so in much the same way as we measure area. Instead of measuring how many 1×1 squares fit, we measure how many $1 \times 1 \times 1$ 'blocks' (or portions of blocks) fit inside the solid.

Finally, we can still use area to measure three-dimensional figures. The **total surface area** of a figure is the area of all of the surfaces that form the borders of the solid. The **lateral surface area** is the total area of all the faces that are not considered 'bases.' We often just ignore lateral surface area and say 'surface area,' which means the same as 'total surface area.'

Problems

Problem 14.4: $ABCDEFGH$ shown is a rectangular prism with $AB = 3$, $BC = 4$, and $AE = 5$.



(a) Find BF and EH.

(b) Find the volume of $ABCDEFGH$.

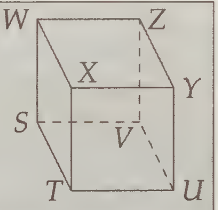
(c) Find the total surface area of $ABCDEFGH$.

(d) Find AC.

(e) What kind of triangle is $\triangle ACG$?

(f) Find AG.

Problem 14.5: A **cube** is a special rectangular prism in which all edges have the same length (i.e., its base is a square and its height has the same length as a side of the base).

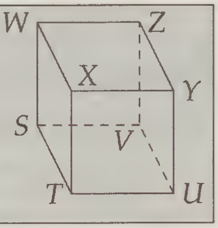


(a) Find the formula for the volume of a cube with side length s.

(b) Find the formula for the total surface area of a cube with side length s.

(c) Find a formula for the length of $\overline{SY}$ in the diagram given that the figure is a cube with side length s.

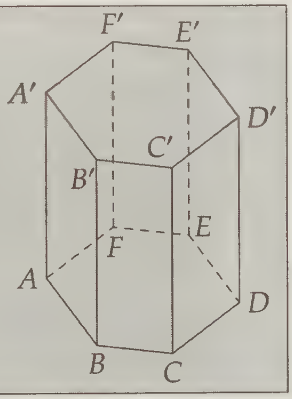
Problem 14.6: Once again, we consider cube $STUVWXYZ$.



(a) What geometric shape is $SXYV$?

(b) What is the area of $SXYV$ if $SV = 4$?

Problem 14.7: The right regular hexagonal prism shown has sides of length 4 on base $ABCDEF$ and a height of 8 units.



(a) Find the lateral surface area of the prism.

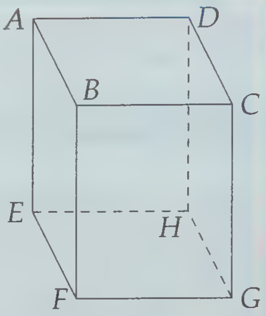
(b) Find the total surface area of the prism.

(c) Find the volume of the prism.

(d) Find AB', AC', and AD' by building the appropriate right triangles.

We start off by finding volumes, areas, and lengths of a (right) rectangular prism.

Problem 14.4: $ABCDEFGH$ shown is a rectangular prism with $AB = 3$, $BC = 4$, and $AE = 5$.



(a) Find BF and EH.

(b) Find the volume of $ABCDEFGH$.

(c) Find the total surface area of $ABCDEFGH$.

(d) Find AC.

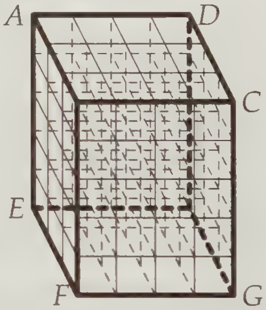
(e) What kind of triangle is $\triangle ACG$?

(f) Find AG.

Solution for Problem 14.4:

(a) All of the faces are rectangles. Since $AEFB$ is a rectangle, $BF = AE = 5$. Similarly, rectangle $EFGH$ tells us that $EH = FG$, and rectangle $BFGC$ gives us $FG = BC = 4$. Therefore, $EH = 4$. We could also have seen that $EH = BC$ by noting that both represent the distance between parallel faces $AEFB$ and $DHGC$.

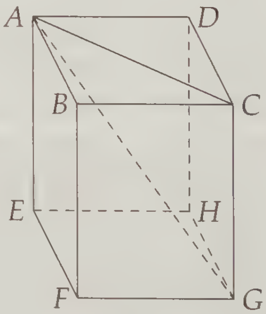
(b) We approach this much as we did finding the area of a rectangle with 1×1 squares on page 85. We can organize a layer of $(AB)(BC) = 12$ $1 \times 1 \times 1$ cubes on face $ABCD$. This plus four more such layers completely fills our prism, so the volume is $3 \times 4 \times 5 = 60$.



(c) To find the total surface area, we need to add the areas of all the faces. Each face is a rectangle, but rather than finding the areas of six different rectangles, we can note that the rectangles come in pairs of congruent rectangles. For example, $ABCD \cong EFGH$. Therefore, our total surface area is:

$$2([ABCD] + [BCGF] + [ABFE]) = 2[(3)(4) + (4)(5) + (3)(5)] = 94.$$

(d) $\overline{AC}$ is a diagonal of rectangle $ABCD$ (or hypotenuse of triangle ABC), so it has length $\sqrt{3^2 + 4^2} = 5$. $\overline{AC}$ is sometimes called a **face diagonal** of the prism.



(e) Since $\overline{CG}$ is perpendicular to plane $ABCD$, it is perpendicular to $\overline{AC}$. Therefore, $\triangle ACG$ is a right triangle.

(f) From right $\triangle ACG$, we have

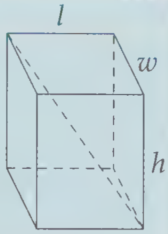
$$AG = \sqrt{GC^2 + AC^2} = \sqrt{25 + 25} = 5\sqrt{2}.$$

Note that $\sqrt{AB^2 + BC^2 + GC^2} = 5\sqrt{2} = AG$. Is this a coincidence?

□

Segment $\overline{AG}$ is called a **space diagonal** of the prism. The lengths AB, AD, and AE are sometimes referred to as the prism's **dimensions**. We might use these to describe the prism as a $3 \times 4 \times 5$ rectangular prism. We can follow the exact same logic we followed in Problem 14.4 to find formulas for the volume, total surface area, and the length of a space diagonal of a rectangular prism.

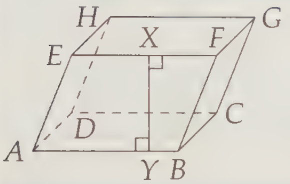
Important: The three dimensions of a rectangular prism are commonly called the length, l, the width, w, and the height, h. For such a prism, we have:



$$\begin{aligned} \text{Volume} &= lwh \\ \text{Surface area} &= 2(lw + wh + lh) \\ \text{Space Diagonal} &= \sqrt{l^2 + w^2 + h^2} \end{aligned}$$

Notice that our volume is simply the area of a base of the prism times the height of the prism. This is true for any prism.

Important: The volume of a prism equals the area of a base times the distance between the bases (i.e. the height).



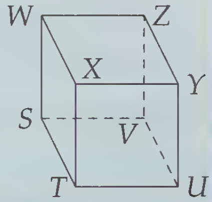
Notice that by 'distance between the bases', we do not necessarily mean the length of the edges connecting corresponding points on the bases. If the prism is not a right prism, then the height is the length of a segment from one base to the other that is perpendicular to both bases. For example, the height between bases $ABCD$ and $EFGH$ of the prism at left is XY, not FB.

Problem 14.5: A **cube** is a special rectangular prism that has all its dimensions the same (i.e., its base is a square and its height has the same length as a side of the base).

(a) Find the formula for the volume of a cube with side length s.

(b) Find the formula for the total surface area of a cube with side length s.

(c) Find a formula for the length of $\overline{SY}$ in the diagram given that the figure is a cube with side length s.

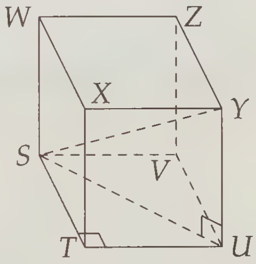


Solution for Problem 14.5:

(a) Since a cube is a rectangular prism, its volume is the product of all three dimensions. Since all three dimensions are the same, s, the volume is simply s^3.

(b) The faces of a cube are all squares. Therefore, each of the 6 faces has area s^2, so the total surface area is $6s^2$.

(c) Once again, we can simply use the formulas we already found for a rectangular prism. But to be a little more sporting, let's prove the formula directly for a cube. To find SY, we'd like to create a right triangle that has $\overline{SY}$ as a side, so we can use the Pythagorean Theorem. Hence, we draw $\overline{SY}$ and $\overline{SU}$, creating right triangle $\triangle SUY$. We have $UY = s$, so all we have to do is find $\overline{SU}$. $\overline{SU}$ is the hypotenuse of $\triangle STU$, so $SU = s\sqrt{2}$. Therefore,



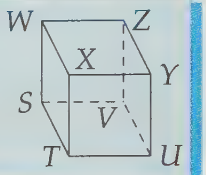
$$SY = \sqrt{SU^2 + UY^2} = \sqrt{2s^2 + s^2} = s\sqrt{3}.$$

□

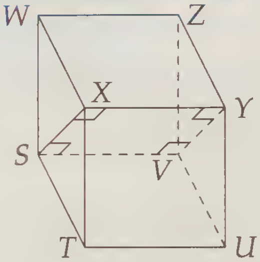
Important: A cube with side length s has:

$$\begin{aligned}\text{Volume} &= s^3\\ \text{Surface area} &= 6s^2\\ \text{Space Diagonal} &= s\sqrt{3}\end{aligned}$$

Problem 14.6: Once again, we consider cube $STUVWXYZ$. What is the area of $SXYV$ if $SV = 4$?



Solution for Problem 14.6: We start by drawing $SXYV$ so we can figure out what sort of shape it is. Since $\overline{SV}$ and $\overline{XY}$ are perpendicular to faces $STXW$ and $VUYZ$, they are both perpendicular to both $\overline{SX}$ and $\overline{YV}$. So, $SXYV$ is a rectangle. We already have one side of the rectangle, $SV = 4$. Therefore, we only need to find SX. Since $\overline{SX}$ is a diagonal of square $SWXT$, $SX = 4\sqrt{2}$, so $[SXYV] = (SV)(SX) = 16\sqrt{2}$. □



One way to view rectangle $SXYV$ of the previous problem is as the intersection of a plane and the cube, where the plane passes through vertices S, X, Y, and V of the cube. We call such an intersection of a plane and a solid a **cross-section** of the solid. Many three-dimensional geometry problems are solved by choosing the right cross-section of the problem to consider. Sometimes we even have to consider multiple cross-sections!

Lest you start to think that all prisms are rectangular ones, we'll try a problem involving a regular hexagonal prism.

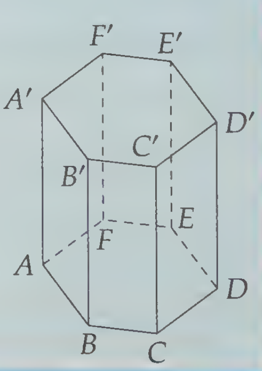
Problem 14.7: The right regular hexagonal prism shown has sides of length 4 on base $ABCDEF$ and a height of 8 units.

(a) Find the lateral surface area of the prism.

(b) Find the total surface area of the prism.

(c) Find the volume of the prism.

(d) Find AB', AC', and AD'.



Solution for Problem 14.7:

(a) Each lateral face of the prism is a 4×8 rectangle. There are 6 lateral faces, so the lateral surface area is $6(4 \times 8) = 192$.

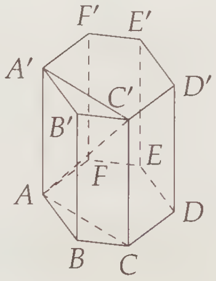
(b) To find the total surface area, we must add the area of the two bases to the lateral surface area. Each base is a regular hexagon. As we saw back in Problem 9.7, we can find the area of a regular hexagon of side length 4 by breaking it into six equilateral triangles with side length 4. Therefore, the area of each base is

$$6\left(\frac{4^2\sqrt{3}}{4}\right) = 24\sqrt{3}.$$

So, the total surface area of the prism is $192 + 2(24\sqrt{3}) = 192 + 48\sqrt{3}$.

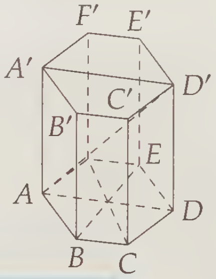
(c) We've already done the hard part! The volume is just the area of a base times the height, or $8(24\sqrt{3}) = 192\sqrt{3}$.

(d) Since $\overline{AB'}$ is a diagonal of rectangle $ABB'A'$, $AB' = \sqrt{4^2 + 8^2} = 4\sqrt{5}$.



The other two are a little trickier. For AC', we consider the cross-section $ACC'A'$ and notice that $\overline{AC'}$ is a diagonal of rectangle $ACC'A'$. We already know $CC' = 8$, so we only have to find AC. Since $\triangle ACD$ is a 30-60-90 triangle (make sure you see why), we have $AC = CD\sqrt{3} = 4\sqrt{3}$. Hence,

$$AC' = \sqrt{AC^2 + CC'^2} = \sqrt{48 + 64} = 4\sqrt{7}.$$



Similarly, $\overline{AD'}$ is a diagonal of rectangle $ADD'A'$. To find AD, we recall the dissection of a regular hexagon by its long diagonals into 6 equilateral triangles as shown, so $AD = 2(BC) = 8$. Therefore,

$$AD' = \sqrt{AD^2 + D'D^2} = \sqrt{64 + 64} = 8\sqrt{2}.$$

□

Concept: Building right triangles works in three dimensions every bit as well as it does in two dimensions.

Exercises

14.2.1 Describe as fully as possible a cross-section of a prism that is parallel to the bases of the prism.

14.2.2 A right rectangular prism has dimensions 2, 5, and $3\sqrt{2}$.

(a) Find the volume of the prism.

(b) Find the total surface area of the prism.

(c) Find the length of a space diagonal of the prism.

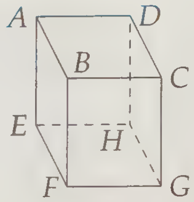
(d) Find the length of the longest face diagonals.

14.2.3 Find the volume of a rectangular prism that has a space diagonal of length 10 and two sides of length 3 and 8.

14.2.4 Find the volume of a cube given that its space diagonal has length 6.

14.2.5 How many different space diagonals does a cube have?

14.2.6 If all the dimensions of a right rectangular prism are doubled, what happens to the surface area? What happens to the volume?



14.2.7 Shown is right rectangular prism $ABCDEFGH$. Given that $AB = 4$, $BC = 3$, and $ABGH$ is a square, find the following:

(a) BD.

(b) AG.

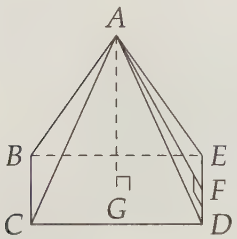
(c) FD.

(d) the volume of $ABCDEFGH$.

14.2.8 Bridget places a box that is $4 \times 6 \times 8$ on the floor. She then places a $2 \times 3 \times 5$ box on top of the first box, forming a two-box tower. She will then paint all the surfaces she can paint without moving either of the boxes. She wants to paint as little as possible, so she places the boxes in a way that minimizes the amount she'll have to paint. What is the total area she has to paint? *(Source: ARML)*

14.3 Pyramids

If we connect all the vertices of a polygon to a point that is not in the same plane as the polygon, we form a **pyramid**. This point is called the **apex** of the pyramid and the polygon is the pyramid's **base**. As we can see at right, the non-base faces of a pyramid are all triangles. The lateral surface area of a pyramid is the sum of the areas of these triangles.

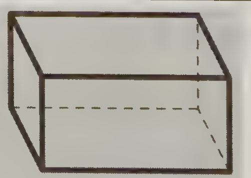


As with prisms, a pyramid is 'regular' if its base is a regular polygon. A pyramid is 'right' if the center of the base is the foot of the altitude from the apex to the base (i.e., the apex is directly 'over' the center of the base). Usually when we speak of a pyramid, we mean a right pyramid. The height of a pyramid is the distance from the apex to the base. For regular pyramids, we also define a **slant height**, which is the distance from the apex to a side of the base. For example, at left, AG is the height and AF is the slant height of the pyramid.

Important: The volume of a pyramid is one-third the product of the pyramid's height and the area of the pyramid's base.

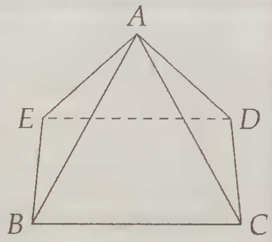
Deriving this formula of a pyramid's volume is very challenging, and will be left for the Intermediate Geometry textbook. However, the formula is not hard to apply, so it is occasionally used in introductory three-dimensional geometry.

Extra! I have two fish tanks that are both rectangular prisms like the one shown to the right. I have a fish that is very picky, and will only be happy in a tank that has exactly 50 gallons of water. Unfortunately, my fish tanks are both 60-gallon tanks. Worse yet, I don't have anything I can use to measure the sides of the tank and mark off a 'fill line.' All I have is a hose to pour the water. How can I fill one of my tanks with exactly 50 gallons?



Problems

Problem 14.8: *ABCDE* is a right square pyramid such that $BC = 6$ and $AC = 5$ as shown at right.



(a) Let O be the center of the square. Why is $\overline{AO}$ perpendicular to the base of the pyramid? Show that $AC = AD$.

(b) Find $[ACD]$ by first finding the length of an altitude from A to $\overline{CD}$.

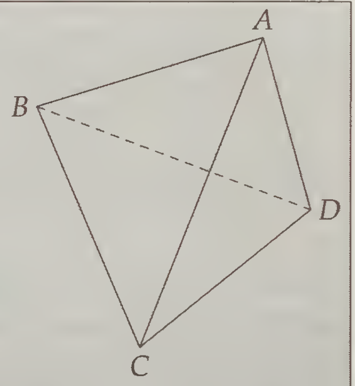
(c) Find the total surface area of *ABCDE*.

(d) Let M be the midpoint of $\overline{CD}$. What kind of triangle is $\triangle AOM$?

(e) Find AO, then find the volume of the pyramid.

Problem 14.9: Find a formula for the lateral surface area of a right pyramid whose base is a regular n-gon with side length s, and whose slant height is l.

Problem 14.10: A triangular pyramid is more commonly called a **tetrahedron**. A **regular tetrahedron** is a tetrahedron whose edges all have the same length. In the diagram, regular tetrahedron *ABCD* has side length 6.



(a) Let G be the foot of the altitude from A to face $\triangle BCD$. What kind of triangles are $\triangle AGB$, $\triangle AGC$ and $\triangle AGD$?

(b) Use part (a) to show that G is the circumcenter of $\triangle BCD$. Why must it also be the centroid of $\triangle BCD$?

(c) Let M be the midpoint of $\overline{CD}$. Find BM and BG.

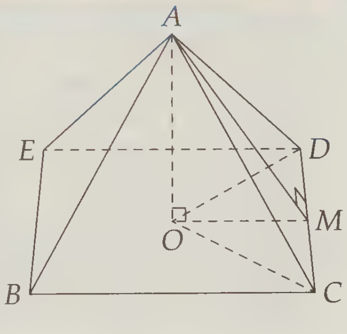
(d) Find AG.

(e) Find the volume of *ABCD*.

We'll start by investigating a square right pyramid.

Problem 14.8: *ABCDE* is a square right pyramid such that $BC = 6$ and $AC = 5$. Find the total surface area and the volume of the pyramid.

Solution for Problem 14.8: We can find the area of the base quickly: $BC^2 = 36$. But to find the area of the triangular faces, we'll need an altitude of one of the faces. We start by drawing $\overline{AO}$, the altitude of the pyramid. Because the pyramid is right, O is the center of base *BCDE*. We suspect that each triangular face is isosceles because the pyramid is right. We can prove this by noting that since $\overline{AO}$ is perpendicular to the base, it is perpendicular to both $\overline{OC}$ and $\overline{OD}$. We have $OC = OD$ because O is the center of the square, so $\triangle AOC \cong \triangle AOD$ by SAS Congruence. This triangle congruence gives us $AC = AD$.



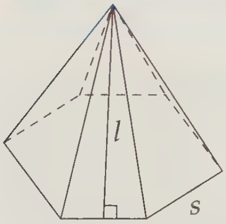
Therefore, $\triangle ACD$ is isosceles, so the median from A to $\overline{CD}$ is also an altitude. Since $AC = 5$ and $CM = CD/2 = 3$, we have $AM = 4$ from right triangle $\triangle AMC$. Therefore, $[ACD] = (AM)(CD)/2 = 12$. Our total surface area consists of four of these triangles plus the square, or $4(12) + 36 = 84$.

Turning to the volume, we need the height of the pyramid, AO. We can build a right triangle by connecting O to either M or to one of the base vertices. In the diagram, we have right triangle $\triangle AOM$. Since $OM = BC/2 = 3$ and $AM = 4$, we have $AO = \sqrt{AM^2 - OM^2} = \sqrt{7}$. Since the area of the base is 36, the volume of the pyramid is $(36)(\sqrt{7})/3 = 12\sqrt{7}$. □

Once again, we built right triangles to find the lengths we needed.

As we saw in that last problem, finding the lateral surface area of a regular right pyramid is easy once we know the slant height and the side length. We can even build a formula for it.

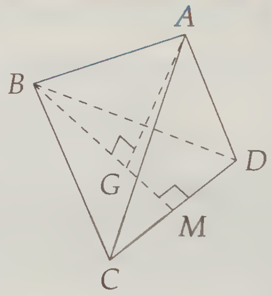
Problem 14.9: Find a formula for the lateral surface area of a right pyramid whose base is a regular n-gon with side length s, and whose slant height is l.



Solution for Problem 14.9: Each of the lateral faces of the pyramid is a triangle with height l and base s. Hence, each of these triangles has area $sl/2$. There is one triangle for each side of the base, so there are n triangles, which have a total area of $n(sl/2) = nsl/2$. Note that ns equals the perimeter of the base, so the lateral surface area of a regular pyramid is half the product of the slant height and the perimeter of the base. □

That formula is so much easier to derive than to memorize that we won't even put it in an 'Important' box.

Problem 14.10: A triangular pyramid is more commonly called a **tetrahedron**. A **regular tetrahedron** is a tetrahedron whose edges all have the same length. Find the volume of a regular tetrahedron that has sides of length 6.



Solution for Problem 14.10: To find the volume, we need an altitude, so we draw altitude $\overline{AG}$ from A to $\triangle BCD$. Since $\overline{AG}$ is perpendicular to plane BCD, triangles $\triangle AGB$, $\triangle AGC$, and $\triangle AGD$ are all right triangles. Because $AB = AC = AD$ and AG is obviously the same in all three triangles, we have $\triangle AGB \cong \triangle AGC \cong \triangle AGD$ by HL Congruence. Therefore, $BG = CG = DG$, which means that G is the circumcenter of $\triangle BCD$ because it is equidistant from the vertices of $\triangle BCD$. Since $\triangle BCD$ is an equilateral triangle, G is also the centroid of $\triangle BCD$.

We can build more right triangles by continuing $\overline{BG}$ to M. Since $\triangle BCD$ is equilateral, $\overline{BM}$ is a median and an altitude. Therefore, $DM = DC/2 = 3$ and $BM = 3\sqrt{3}$ (from 30-60-90 $\triangle BMD$). Since the centroid of a triangle divides its medians in a 2 : 1 ratio, we have $BG = (2/3)BM = 2\sqrt{3}$. Finally, we can find AG from right triangle $\triangle AGB$:

$$AG = \sqrt{AB^2 - BG^2} = \sqrt{36 - 12} = 2\sqrt{6}.$$

Since the area of $\triangle BCD$ is $(DC)(BM)/2 = 9\sqrt{3}$, our volume is $([BCD])(AG)/3 = 18\sqrt{2}$.

Similarly, we can show that the volume of a regular tetrahedron with edge length s is $s^3\sqrt{2}/12$. □

Important: Problem 14.10 is a typical challenging three-dimensional geometry problem in that our general tactic is to reduce it to a series of two-dimensional problems. When you can work through this problem on your own, you're ready for some serious three-dimensional geometry.

Exercises

14.3.1 $ABCDE$ is a right square pyramid with base $ABCD$. Given that $AB = 4$ and $AE = 8$, find the following:

(a) The height of the pyramid.

(b) The slant height of the pyramid.

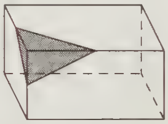
(c) The volume of the pyramid.

(d) The total surface area of the pyramid.

14.3.2 When we first created Problem 14.8, we let $BC = 8$ instead of $BC = 6$ in the problem statement. Why did we have to change it? **Hints:** 477

14.3.3 A regular pyramid with a square base has base edges of length 6 inches and height 4 inches. What is the ratio of the number of cubic inches in the volume of the pyramid to the number of square inches in its surface area? *(Source: MATHCOUNTS)*

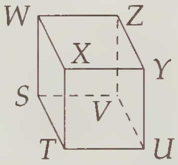
14.3.4 A $4'' \times 6'' \times 8''$ rectangular solid is cut by slicing through the midpoints of three adjacent sides as shown.



(a) Find the volume of the shaded piece that is cut off.

(b) Find the sum of the lengths of the edges of the shaded piece that is cut off. *(Source: MATHCOUNTS)*

14.3.5 $STUVWXYZ$ is a cube as shown with $ST = 4$.



(a) What are the volume and surface area of pyramid $STUW$?

(b) What are the volume and surface area of pyramid $STUX$? **Hints:** 524

(c) What are the volume and surface area of pyramid $STUZ$?

Extra! Back on page 253, we noted that any polygon can be dissected and rearranged to form any other polygon that has the same area. You might wonder if it is possible to dissect any polyhedron and rearrange the pieces to form any other polyhedron with the same volume. You wouldn't be the first to wonder this! In fact, this question was one of Hilbert's famous problems (see page 56). It was also the first to be solved, by **Max Wilhelm Dehn**.

Dehn used the powerful problem solving technique of **invariants** to solve the problem. He created a function of the edge lengths and the dihedral angles of a polyhedron and showed that this function must be equal for any two polyhedra that can be dissected into the same pieces. The function is different for a cube and a tetrahedron, so these two cannot be dissected into the same pieces.

14.4 Regular Polyhedra

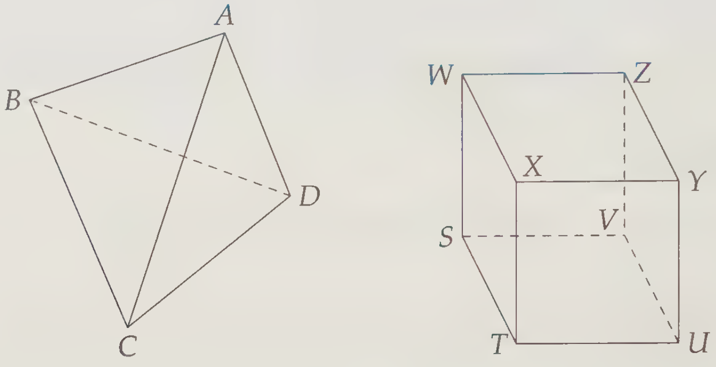


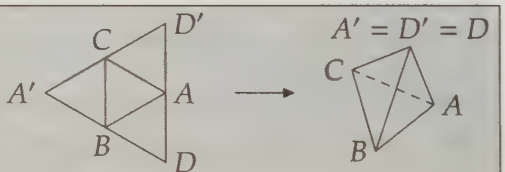
Figure 14.2: A Regular Tetrahedron and a Cube

We have already mentioned that a polyhedron is a solid figure that is bounded on all sides by polygons. A **regular polyhedron** is a convex polyhedron in which all the faces are congruent regular polygons, and there are the same number of edges at each vertex. We've already seen examples of two such regular polyhedra – a cube and a regular tetrahedron, examples of which are shown in Figure 14.2. The faces of the cube are congruent squares and those of the tetrahedron are congruent equilateral triangles.

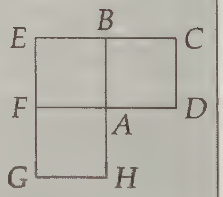
These aren't the only types of regular polyhedra, however. In this section we discover the other regular polyhedra, and we tackle a problem involving one of the types of regular polyhedra.

Problems

Problem 14.11: To make a regular tetrahedron, we might start with four equilateral triangles in a plane as shown at right. We then fold the triangles until A', D, and D' coincide as shown in the regular tetrahedron. As we'll see in this problem, thinking about 'folding up' regular polygons like this allows us to limit the possibilities of regular polyhedra that can be made.



(a) Consider three squares, $ABCD$, $ABEF$, and $AFGH$, arranged around a point as shown. If we 'fold' this arrangement so that D and H coincide, we'll be on our way to forming what type of regular polyhedron?



(b) Can we fit three regular pentagons around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron?

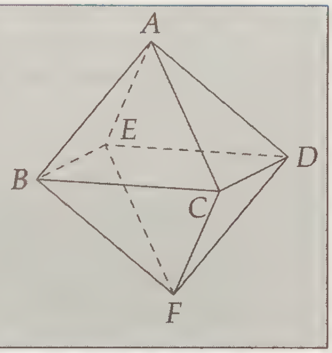
(c) Can we fit three regular hexagons around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron?

(d) Can we fit four equilateral triangles around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron? How about four squares?

(e) Can we fit five equilateral triangles around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron? How about five squares?

(f) At most how many different types of regular polyhedra are there?

Problem 14.12: $ABCDEF$ shown is a regular octahedron, which has eight congruent equilateral triangles as faces. We can also think of a regular octahedron as a pair of square pyramids glued together at their bases. In this problem, let $AB = 4$.



(a) Find BD.

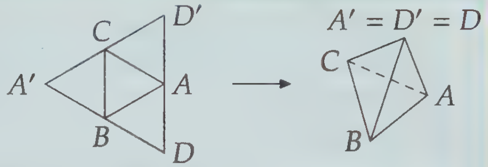
(b) Find AF.

(c) Find $[BCDE]$.

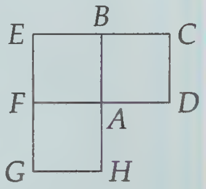
(d) Find the volume of $ABCDEF$.

There aren't many types of regular polyhedra. With a little physical intuition, we can discover which types of polyhedra might possibly exist.

Problem 14.11: To make a regular tetrahedron, we might start with four equilateral triangles in a plane as shown at right. We then fold the triangles until A, D, and D' coincide as shown in the regular tetrahedron. As we'll see in this problem, thinking about 'folding up' regular polygons like this allows us to limit the possibilities of regular polyhedra that can be made.



(a) Consider three squares, $ABCD$, $ABEF$, and $AFGH$, arranged around a point as shown. If we 'fold' this arrangement so that D and H coincide, we'll be on our way to forming what type of regular polyhedron?



(b) Can we fit three regular pentagons around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron?

(c) Can we fit three regular hexagons around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron?

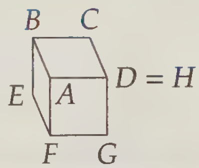
(d) Can we fit four equilateral triangles around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron? How about four squares?

(e) Can we fit five equilateral triangles around a point in a plane? Can we then fold the resulting figure to start to form a polyhedron? How about five squares?

(f) At most how many different types of regular polyhedra are there?

Solution for Problem 14.11:

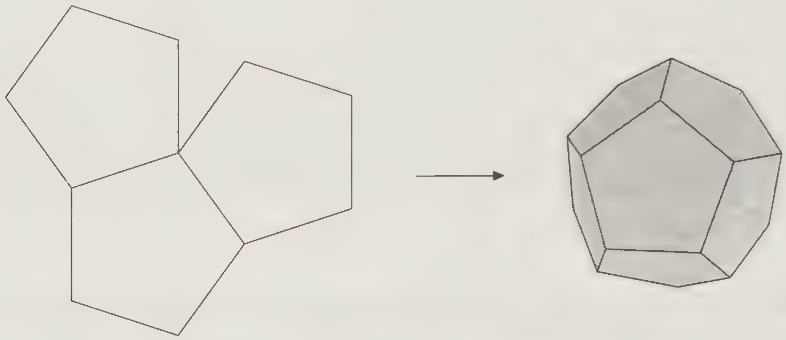
(a) Folding up our squares gives the figure shown, which is clearly well on its way to being a cube.



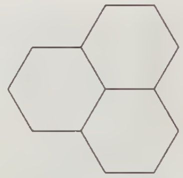
Extra! *We have to reinvent the wheel every once in a while, not because we need a lot of wheels, but because we need a lot of inventors.*

–Bruce Joyce

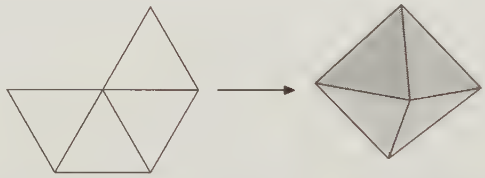
(b) We can fit three pentagons around a point with room to spare. Therefore, we can fold the pentagons to possibly form part of a regular polyhedron. Such a polyhedron does indeed exist, as shown below. This polyhedron is called a regular **dodecahedron**. It has 12 regular pentagons as its faces.



(c) We can fit three regular hexagons around a point, but since each angle of a regular hexagon is 120°, the three fit snugly. In other words, we can't bend the hexagons together as we've done earlier with the pentagons, squares and triangles. Therefore, there are no more regular polyhedra that have three of each polygon meeting at a point.

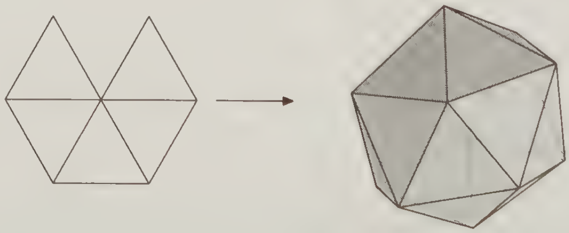


(d) Four equilateral triangles fit around a point with room to spare, so we can fold them together. The result is a square pyramid without its base. Gluing another square pyramid to this one gives us a polyhedron with eight congruent equilateral triangles as faces. This polyhedron, shown in the diagram below, is a regular **octahedron**.



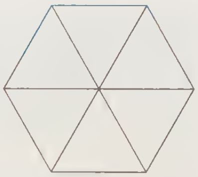
Four squares, of course, fit snugly around a point, since $4(90°) = 360°$. Hence, we can't fold the squares to make a polyhedron.

(e) As with three and four equilateral triangles, we can fit five triangles around a point, then fold them together. This gives us the start of a regular **icosahedron**, which has twenty congruent equilateral triangles as its faces.



Obviously, we can't fit five of any other regular polygon around a point.

(f) We can fit six equilateral triangles snugly around a point, so we can't 'fold' them into the start of a regular polyhedron. Obviously, any regular polyhedron must have at least three faces meet at each vertex, so we've exhausted all the possibilities for regular polyhedra.



□

Important: There are five regular polyhedra, which are described below.

Name	Face Type	# Faces	# Edges	# Vertices
Tetrahedron	Triangle	4	6	4
Cube	Square	6	12	8
Octahedron	Triangle	8	12	6
Dodecahedron	Pentagon	12	30	20
Icosahedron	Triangle	20	30	12

These five regular polyhedra are collectively named the **Platonic solids**, after the great Greek philosopher **Plato**.

Icosahedra and dodecahedra are pretty hard to work with, but since regular octahedra are just two square pyramids stuck together, they're much more manageable.

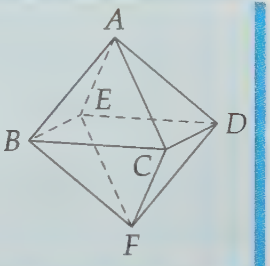
Problem 14.12: $ABCDEF$ shown is a regular octahedron with $AB = 4$.

(a) Find BD.

(b) Find AF.

(c) Find $[BCDE]$.

(d) Find the volume of $ABCDEF$.



Solution for Problem 14.12:

(a) Since $BCDE$ is a square, $BD = BC\sqrt{2}$. All the sides of a regular polyhedron have the same length, so $BC = AB = 4$ and $BD = 4\sqrt{2}$.

(b) Instead of thinking of the octahedron as square pyramids $ABCDE$ and $FBCDE$ glued together, we can think of it as $BAEFC$ and $DAEFC$ glued together. Therefore, $ACFE$ is also a square, so $AF = BD = 4\sqrt{2}$. (We can also just note that since the polyhedron is regular, the distance between directly opposite vertices must be the same no matter which pair of opposite vertices we choose.)

(c) $BCDE$ is a square with side length 4, so its area is $4^2 = 16$.

(d) To find the volume of the regular octahedron, we think of it as two congruent square pyramids, $ABCDE$ and $FBCDE$. Since the altitudes from A and from F, respectively, of these two pyramids together make up $\overline{AF}$, the height of each pyramid is $AF/2 = (4\sqrt{2})/2 = 2\sqrt{2}$. The area of a base is $4^2 = 16$, so the volume of each pyramid is $(16)(2\sqrt{2})/3 = 32\sqrt{2}/3$. Therefore, our octahedron has volume $2(32\sqrt{2}/3) = 64\sqrt{2}/3$.

□

Exercises

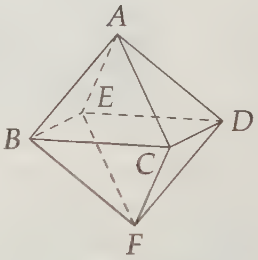
14.4.1 Compute the quantity

$$(\text{Number of vertices}) - (\text{Number of edges}) + (\text{Number of faces})$$

for each regular polyhedron (cube, tetrahedron, octahedron, dodecahedron, icosahedron). Notice anything interesting? Try it on other polyhedra, such as various types of prisms and pyramids and see if your interesting observation still holds.

14.4.2 The six vertices of a regular octahedron are snipped off, leaving a square face in place of each corner and a hexagonal face in place of each original face of the octahedron. How many vertices, faces, and edges will the new polyhedron have?

14.4.3★ $ABCDEF$ shown at right is a regular octahedron with side length 1. Let O be the center of face ABC, P be the center of face ACD, and Q be the center of face DEF.



(a) Find OP. **Hints:** 287

(b)★ Find OQ. **Hints:** 59

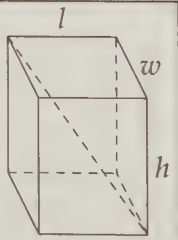
(c) Find $[OPQ]$. **Hints:** 580

14.5 Summary

Definitions: The **volume** of a three dimensional figure is a measure of the space inside the figure. The **total surface area** of a figure is the total area of all the surfaces that form a boundaries of the solid. The **lateral surface area** is the total area of all the surfaces that are not considered 'bases'.

Definitions: A **polyhedron** is a solid figure with polygons as its boundaries. A **prism** has two congruent parallel faces as **bases** and all remaining faces (called **sides**) are parallelograms. In a **right prism** all of these side faces are rectangles. The bases are used to describe the prism, as in 'right rectangular prism' (shown below) or 'hexagonal prism'.

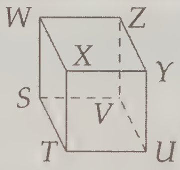
Important: The three dimensions of a right rectangular prism are commonly called the length, l, the width, w, and the height, h. For such a prism, we have:



$$\begin{aligned} \text{Volume} &= lwh \\ \text{Surface area} &= 2(lw + wh + lh) \\ \text{Space Diagonal} &= \sqrt{l^2 + w^2 + h^2} \end{aligned}$$

The volume of a prism equals the area of the base times the distance between the bases (i.e. the height).

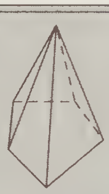
Definition: A **cube** is a special right rectangular prism in which all the edge lengths are the same (i.e., its base is a square and its height has the same length as a side of the base).



Important: A cube with side length s has:

$$\begin{aligned} \text{Volume} &= s^3 \\ \text{Surface area} &= 6s^2 \\ \text{Space Diagonal} &= s\sqrt{3} \end{aligned}$$

Definitions: If we connect all the vertices of a polygon to a point that is not in the same plane as the polygon, we form a **pyramid**. This point is called the **apex** of the pyramid and the polygon is the pyramid's **base**. As we can see at right, the non-base faces of a pyramid are all triangles. The lateral surface area of a pyramid is the sum of the areas of these triangles. A **tetrahedron** is a pyramid with a triangular base.



The **height** of a pyramid is the distance from the apex to the base. If the base is a regular polygon, the pyramid is a **regular pyramid**. For regular pyramids, we also define a **slant height**, which is the distance from the apex to a side of the base.

Important:

- The volume of a pyramid is one-third the product of the pyramid's height and the area of the pyramid's base.
- The lateral surface area of a regular pyramid equals one-half the product of the slant height and the perimeter of the pyramid's base.

Definitions: A **regular polyhedron** is a polyhedron whose faces are all congruent regular polygons.

Important: There are five regular polyhedra, which are described below.

Name	Face Type	# Faces	# Edges	# Vertices
Tetrahedron	Triangle	4	6	4
Cube	Square	6	12	8
Octahedron	Triangle	8	12	6
Dodecahedron	Pentagon	12	30	20
Icosahedron	Triangle	20	30	12

Problem Solving Strategies

Concept:

- Building right triangles works in three dimensions every bit as well as it does in two dimensions.

REVIEW PROBLEMS

14.13 A diagonal of one of the faces of a given cube has length 4.

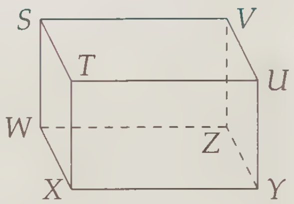
(a) Find the length of an edge of the cube.

(b) Find the length of a space diagonal of the cube.

(c) Find the total surface area of the cube.

(d) Find the volume of the cube.

14.14 Shown is right rectangular prism $ZYXWVUTS$. What type of quadrilateral is $ZYTS$ and why?



14.15 A right rectangular prism has space diagonal $3\sqrt{13}$ and two sides of length 3 and 7, respectively.

(a) Find the third dimension of the prism.

(b) Find the volume of the prism.

(c) Find the total surface area of the prism.

14.16 A space diagonal of cube $\mathcal{A}$ is an edge of cube $\mathcal{B}$.

(a) Find the ratio of the surface area of $\mathcal{A}$ to the surface area of $\mathcal{B}$.

(b) Find the ratio of the volume of $\mathcal{A}$ to the volume of $\mathcal{B}$.

14.17 $VWXYZ$ is a right square pyramid with square base $WXYZ$. Given $YZ = 10$ and $YV = 13\sqrt{2}$, find the following:

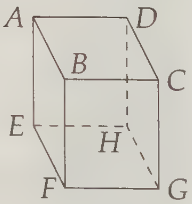
(a) the height of the pyramid.

(b) the slant height of the pyramid.

(c) the total surface area of the pyramid.

(d) the volume of the pyramid.

14.18 $M, N, O,$ and P are midpoints of edges $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$, respectively, of right rectangular prism $ABCDEFGH$ shown. Q, R, S, and T are the midpoints of edges $\overline{EF}$, $\overline{FG}$, $\overline{GH}$, and $\overline{HE}$, respectively. Given $AB = 8$, $BC = 10$, and $BF = 5$, find the following:



(a) MN.

(b) MO.

(c)★ the volume of $MNOPQRST$.

14.19 What figure is formed by connecting the centers of the faces of a:

(a) cube?

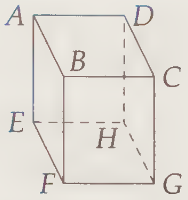
(b) regular octahedron?

(c) regular tetrahedron?

(d) regular dodecahedron?

(e) regular icosahedron?

14.20 Shown is cube $ABCDEFGH$ with side length 8.

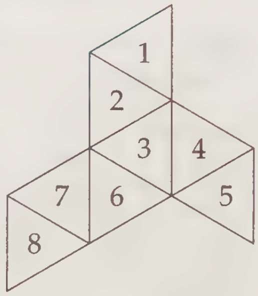


(a) Find $[ACF]$. **Hints:** 332

(b) Find $[ACG]$.

14.21 How many edges does a regular hexagonal prism have?

14.22 This shape at right is folded into a regular octahedron with the equilateral triangles shown as faces. What is the sum of the numbers on the faces sharing an edge with the face with a "1" on it? *(Source: MATHCOUNTS)*



14.23 $ABCDEFG$ is a right pyramid with regular hexagonal base $ABCDEF$ and apex G with $AB = 6$ and $AG = 6\sqrt{3}$.

(a) Find the volume of $ABCDEFG$.

(b) Find the total surface area of $ABCDEFG$.

14.24 Find the volume and the surface area of a regular tetrahedron with side length 9.

14.25 On the center of each face of a cube with side length 4 inches, we glue a cube of side length 1 inch (so that an entire face of each small cube is glued to the big cube). We then paint the entire resulting figure red. How many square inches of red paint will we use?

14.26 $WXYZ$ is a triangular pyramid with $XY = YZ = ZX = 9$ and $WX = WY = WZ = 18$. Let G be the centroid of $\triangle XYZ$ and M the midpoint of $\overline{XY}$.

(a) Prove that $\overline{WG}$ is perpendicular to plane XYZ. (If you're stuck, check out Problem 14.10.)

(b) Find WM.

(c) Find ZM.

(d) Find the volume of $WXYZ$.

14.27 In a cube with side length 6, what is the volume of the tetrahedron formed by any vertex and the three vertices connected to that vertex by edges of the cube? *(Source: HMMT)*

14.28 Each of the three lines k, m, and n is perpendicular to the other two lines. All three lines pass through point P. K is on line k, M is on line m, and N is on line n such that $KP = 7$, $KM = 9$, and $KN = 11$.

(a) Find the length of $\overline{MN}$.

(b) Find the volume of $KNMP$.

(c)★ Find the length of the segment from K to the midpoint of $\overline{MN}$. **Hints:** 554, 371

14.29 In tetrahedron $MNOP$, we have $MN = OP$, $MO = NP$, and $MP = NO$. Prove that $\angle MNO = \angle MPO$.

14.30 Find a formula for the volume of a regular octahedron with edge length s.

Challenge Problems

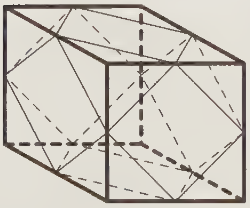
14.31 Base $EFGH$ of right prism $EFGHIJKL$ is a parallelogram with $EF = 8$, $FG = 6$, and $\angle EFG = 60^\circ$. Given that the height of the prism is 9, find the following:

(a) $[EFGH]$. **Hints:** 542

(b) the surface area of $EFGHIJKL$.

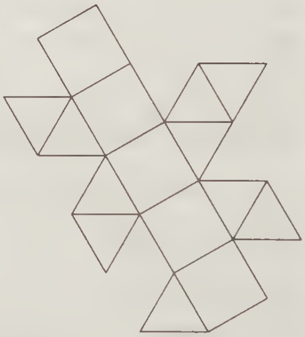
(c) the volume of $EFGHIJKL$.

14.32 The altitude from vertex V of tetrahedron $VWXY$ meets face WXY at the circumcenter of $\triangle WXY$. Prove that $VW = VX = VY$. **Hints:** 425



14.33 Each side of the cube shown has length 10. What is the volume of the solid whose edges are formed by connecting all the midpoints of the edges of the cube as shown? **Hints:** 50

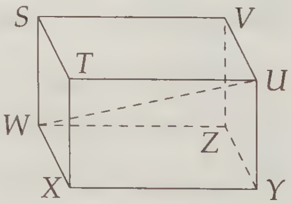
14.34 A soccer ball is constructed using 32 regular polygons with equal side lengths. Twelve of the polygons are pentagons, and the rest are hexagons. A seam is sewn wherever two edges meet. What is the number of seams in the soccer ball? *(Source: MATHCOUNTS)* **Hints:** 130



14.35 The figure at right with 5 square faces and 10 equilateral triangular faces is folded into a 15-faced polyhedron. How many edges does the polyhedron have? How many vertices does the polyhedron have? *(Source: MATHCOUNTS)*

14.36 An octahedron is formed by connecting the centers of the faces of a cube. What is the ratio of the volume of the cube to the volume of the octahedron? **Hints:** 164

14.37 An insect lives on the surface of a regular tetrahedron with edges of length 1. It wishes to travel on the surface of the tetrahedron from the midpoint of one edge to the midpoint of the opposite edge. What is the length of the shortest such trip? (Note: Two edges of a tetrahedron are opposite if they have no common endpoint.) *(Source: AMC 12)* **Hints:** 111



14.38 $STUVWXYZ$ is a right rectangular prism with $XY = 12$, $WX = 3$, and $TX = 4$.

(a) Find WU.

(b) Find the distance from Z to $\overline{WU}$. **Hints:** 64, 531

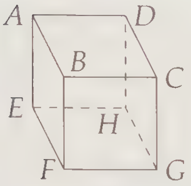
(c) Find the volume of $TWXY$.

(d)★ Find the volume of pyramid $VSUYW$. (V is the vertex.)

(e)★ Find the volume of $WXZU$. **Hints:** 155, 573

14.39 A cube of cheese $C = \{(x, y, z) | 0 \le x, y, z \le 1\}$ is cut along the planes $x = y$, $y = z$, and $z = x$. How many pieces are produced this way? *(Source: AHSME)* **Hints:** 181, 253

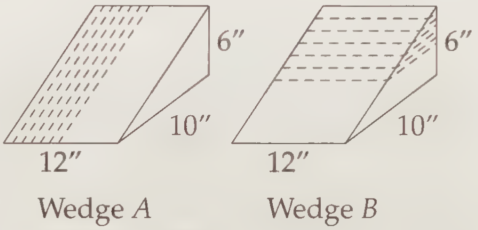
14.40 *EFGHIJKL* is a cube. How many different planes pass through at least three vertices of *EFGHIJKL*? **Hints:** 297



14.41 In cube *ABCDEFGH* shown, $AB = 12$. Find the distance from B to plane ACF. **Hints:** 93, 281

14.42 The sum of the lengths of the twelve edges of a rectangular box is 140, and the distance from one corner of the box to the farthest corner is 21. What is the total surface area of the box? *(Source: AMC 12)* **Hints:** 431

14.43 Congruent wedges A and B are sliced as shown. Each wedge has a pair of parallel triangular faces. Wedge A is cut vertically so that slices are made perpendicular to the base; wedge B is cut horizontally so that slices are made parallel to the base. Six slices 1/4 inch thick are cut from each wedge. What is the ratio of the remaining volume of wedge A to the remaining volume of wedge B? *(Source: MATHCOUNTS)* **Hints:** 45



14.44 *ABCDEF* shown at left below is a regular octahedron with edge length 1. The midpoints of edges $\overline{AB}$, $\overline{AC}$, $\overline{AD}$, and $\overline{AE}$ are M, N, O, and P, respectively. The midpoints of edges $\overline{BF}$, $\overline{CF}$, $\overline{DF}$, and $\overline{EF}$ are Q, R, S, and T, respectively. Find the volume of *MNOPQRST*. **Hints:** 96

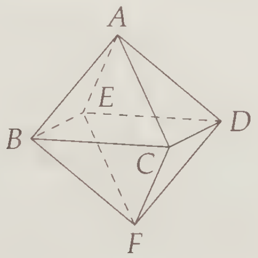


Figure 14.3: Diagram for Problem 14.44

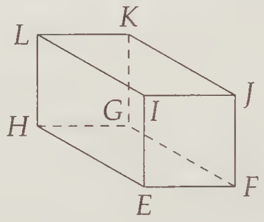


Figure 14.4: Diagram for Problem 14.45

14.45★ Parallelogram *EFGH* is a base of right prism *EFGHIJKL* at right above. Given $EF = 8$, $FG = 6$, $EI = 9$, and $\angle EFG = 60^\circ$, find FL and EK. **Hints:** 133

14.46★ A right square **frustum** is formed by cutting a right square pyramid with a plane parallel to its base. Suppose the original pyramid has base length 6 and height 9, and that the plane cutting the pyramid to form the frustum is 3 units from the base of the pyramid.

(a) Find the volume of the frustum. **Hints:** 185, 263

(b) Find the total surface area of the frustum.

14.47★ Let *ABCD* be a regular tetrahedron with side length 2. The plane parallel to edges $\overline{AB}$ and $\overline{CD}$ and lying halfway between them cuts *ABCD* into two pieces. Find the surface area of one of these pieces. *(Source: HMMT)* **Hints:** 211